

# Osprey Rudder — Engineering Constraints

[T] thesis · [SIM] simulation ( python -m osprey size ) · [TEAM] team decision · **bold** = unverified

**Existing rudder.** MHZ Mystic C5000, 150 × 30 mm blade, about **3 in (76 mm)** **submerged** on plane (unmeasured). Mount is as low as the transom allows [T §3.4.2] . AGFRC A81FHM HV servo, **74 kg-cm stall at 8.4 V**, pull-pull cables [T §3.4.1] . **Race plan:** 50 mph cruise [TEAM] .



Installed: transom mount, rudder block atop the blade for the cable actuators. Thesis Fig 5.6.

## Constraints

| #  | CONSTRAINT                                                                 | REASON                                                                                                                                            |
|----|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 | <b>Span buys authority; chord mostly buys drag and hinge moment</b>        | Authority is $A_r \cdot CL_a$ . At 5 in span, going 30 → 45 mm chord adds 23% authority for +50% area, +51% drag and 1.8× the hinge moment. [SIM] |
| R2 | <b>Useful deflection ±10°</b> — autopilot clamps here, not at the stop     | Blade ventilates near 10°; moment then drops ~40% and stays down until ~6°. [SIM]                                                                 |
| R3 | <b>Mechanical stop ±35° — assumed</b>                                      | Set by horn and mount tabs. Measure.                                                                                                              |
| R4 | <b>Stock at 20–25% chord aft of LE</b>                                     | Hinge moment is $F \cdot (x_{cp} - x_{stock}) \cdot c$ . At 45 mm chord this decides whether the servo works at all (below).                      |
| R5 | <b>Mount stays; blade extends 2 in downward</b>                            | Mount cannot go below the transom [T §3.4.2] . Block and cables unchanged.                                                                        |
| R6 | <b>Model valid to 44 mph</b>                                               | Cavitation number < 0.5 above 19.9 m/s. The 50 mph race is past this: extrapolated. [SIM]                                                         |
| R7 | <b>Autopilot steers with motors below 10 mph, rudder only above</b> [TEAM] | Rudder authority goes as $U^2$ ; at launch it has almost none.                                                                                    |

## Spec — 45 mm × 5 in [TEAM]

|                     | CURRENT             | NEW                       |                            | CURRENT         | NEW                   |
|---------------------|---------------------|---------------------------|----------------------------|-----------------|-----------------------|
| Submerged span      | 3.0 in / 76 mm      | <b>5.0 in / 127 mm</b>    | Total blade, same mount    | 5.9 in          | <b>7.9 in</b>         |
| Chord               | 30 mm               | <b>45 mm</b>              | Stock position             | <b>unknown</b>  | <b>20–25% c</b>       |
| Wetted area         | 3.5 in <sup>2</sup> | <b>8.9 in<sup>2</sup></b> | Authority $A_r \cdot CL_a$ | 1.0×            | <b>2.7×</b>           |
| Aspect ratio        | 2.5                 | <b>2.8</b>                | Turn radius at 10°, 5 m/s  | 28 m (13.3 LOA) | <b>19 m (8.9 LOA)</b> |
| Drag at 10°, 10 m/s | 4.3% of hull        | <b>11.0%</b>              | Peak rudder, 50 mph race   | 7.9°            | <b>6.5°</b>           |

## Servo check, new blade — 74 kg-cm stall

| SPEED           | STOCK AT LE      | 25% C, COP 0.10C OFF | 25% C, COP NOMINAL |
|-----------------|------------------|----------------------|--------------------|
| 30 mph          | 37 / 47 kg-cm    | 15 / 19              | ~0                 |
| <b>50 mph †</b> | <b>103 / 132</b> | <b>41 / 53</b>       | ~0                 |
| 70 mph †        | 202 / 258        | 81 / 103             | ~0                 |

Worst case from straight running: travel clamped at 10° / full travel to 35°. With full travel the peak is *past* ventilation onset (~26°), not at it. † past R6. In the simulated 50 mph race the autopilot needed **under 10 kg-cm**; the table is the step-input worst case. **Verdict: the servo is adequate at 50 mph only with the stock near 25% chord.** With an LE stock, use a 2:1 tiller arm or a stronger servo.

The thesis's 66.8 kg-cm (drag × horn radius, 80 mph) is *not* conservative: the correct hinge moment for the old blade with an LE stock is 65 clamped / 83 full travel [SIM] . It lands close by coincidence.

**Measure before building:** submerged depth on plane · stock position · tiller arm radius · mechanical stop · trailer and ramp draft with 2 in more blade.